

Literature Review on Etripamil: A Novel FDA-Approved Intranasal Calcium Channel Blocker for Paroxysmal Supraventricular Tachycardia (PSVT)

Abstract

Cardiovascular diseases remain one of the leading causes of morbidity worldwide, creating a continuous need for innovative therapeutic approaches. Paroxysmal supraventricular tachycardia (PSVT) is a common cardiac arrhythmia characterized by sudden episodes of abnormally rapid heartbeat originating above the ventricles. Conventional management of acute PSVT episodes mainly depends on vagal maneuvers or intravenous drugs such as adenosine, diltiazem, and verapamil administered under medical supervision.

Etripamil, marketed as Cardamyst™, represents an important advancement in cardiovascular therapy as a novel intranasal, self-administered L-type calcium channel blocker developed for rapid conversion of PSVT episodes to normal sinus rhythm. It was designed to overcome limitations associated with hospital-based treatment by allowing patients to manage episodes outside healthcare settings. The FDA approval of etripamil was supported by extensive clinical evaluation, including the Phase III RAPID study program demonstrating significant efficacy and acceptable safety.

This review focuses on the discovery, development, clinical trials, therapeutic applications, and market impact of etripamil.

Keywords: Etripamil, Cardamyst, PSVT, calcium channel blocker, cardiovascular drug, intranasal drug delivery

1. Introduction

Paroxysmal supraventricular tachycardia is a rhythm disorder involving sudden episodes of rapid heart rate caused by abnormal electrical conduction pathways above the ventricles. Patients commonly experience palpitations, chest discomfort, anxiety, dizziness, fatigue, and shortness of breath. Although many episodes are not immediately life-threatening, recurrent attacks significantly affect quality of life and frequently result in emergency healthcare visits.

Traditional pharmacological management involves intravenous administration of antiarrhythmic agents, particularly adenosine and calcium channel blockers. However, these treatments require trained healthcare professionals and clinical monitoring. This created a therapeutic gap for a rapid, convenient, patient-controlled treatment.

Etripamil was developed to address this limitation through a nasal spray formulation capable of rapid absorption and short duration of action. It provides an on-demand treatment approach, representing a shift from hospital-dependent therapy toward patient-managed arrhythmia care.

2. Drug Discovery and Development Process

Etripamil was developed by Milestone Pharmaceuticals as a novel short-acting calcium channel blocker specifically optimized for intranasal delivery. Unlike conventional calcium channel blockers designed for oral or intravenous administration, etripamil was engineered to produce rapid therapeutic effects with limited systemic exposure.

The major target of etripamil is the L-type calcium channel present in cardiac tissues. Calcium influx through these channels plays an essential role in atrioventricular (AV) nodal conduction. By blocking calcium entry, etripamil temporarily slows AV nodal conduction and interrupts re-entry circuits responsible for PSVT.

The development strategy focused on several important goals:

- Rapid onset of action
- Easy self-administration
- Short pharmacological duration
- Reduction in emergency medical intervention
- Improved patient independence

The nasal route was selected because the nasal mucosa contains rich vascular supply, allowing quick drug absorption while avoiding first-pass hepatic metabolism.

Etripamil represents an example of patient-centered drug development where an existing pharmacological principle (calcium channel blockade) was combined with an innovative delivery system.

3. Clinical Trial Development

Clinical evaluation of etripamil involved multiple studies assessing safety, dose optimization, and therapeutic efficacy.

Phase I Clinical Trials

The initial Phase I studies evaluated:

- Safety profile
- Pharmacokinetics
- Pharmacodynamics
- Dose tolerance

Healthy volunteers were administered different doses of intranasal etripamil to determine its effect on cardiovascular parameters.

The results demonstrated rapid absorption and short-lasting effects, supporting further clinical investigation.

Phase II Clinical Trials

Phase II trials investigated etripamil in patients experiencing PSVT.

Objectives included:

- Determination of effective dose
- Evaluation of conversion from PSVT to sinus rhythm
- Assessment of adverse effects

These studies helped establish the 70 mg intranasal dose selected for advanced clinical development.

Phase III Clinical Trials: RAPID Study

The RAPID trial was a major randomized, double-blind, placebo-controlled Phase III clinical study evaluating etripamil nasal spray in PSVT patients.

Study characteristics:

Parameter	Details
Trail	Rapid study
Disease	PSVT
Dose	70mg Intranasal etripamil
Design	randomized, double-blind, placebo controlled
Primary endpoint	conversion of PSVT to sinus rhythm

The study demonstrated that etripamil significantly improved conversion of PSVT episodes compared with placebo. Approximately 64% of patients receiving etripamil converted to sinus rhythm within 30 minutes compared with approximately 31% receiving placebo.

Common adverse effects were mainly related to nasal administration, including:

- Nasal discomfort
- Nasal congestion
- Rhinorrhea
- Throat irritation

Overall results supported the safety and efficacy of patient-administered etripamil therapy.

4. Therapeutic Applications

The primary therapeutic application of etripamil is:

- Treatment of acute symptomatic episodes of paroxysmal supraventricular tachycardia in adults.
 - Intranasal delivery provides quick absorption, helping restore normal rhythm during acute episodes.
 - Patients can administer the drug without immediate hospital assistance, improving convenience.
 - Effective home treatment may decrease emergency department visits.
 - Availability of immediate therapy reduces anxiety associated with unpredictable PSVT attacks.
-

5. Market Impact

The approval of etripamil introduced an important change in the cardiovascular drug market by creating a new category of self-administered arrhythmia treatment.

Important market factors include:

- First-in-class advantage
- Patient convenience
- Healthcare cost reduction

Commercial potential

Millions of patients experience PSVT worldwide, creating a significant market opportunity for convenient therapies.

The successful development of etripamil may encourage future research into patient-controlled treatments for cardiovascular emergencies.

Conclusion

Etripamil represents a significant advancement in cardiovascular pharmacotherapy by combining established calcium channel blockade with innovative intranasal drug delivery. Its development addressed an unmet need in PSVT treatment by allowing rapid, self-administered management of arrhythmia episodes.

Clinical trials demonstrated its ability to effectively restore sinus rhythm with a favorable safety profile. The FDA approval of etripamil highlights the importance of developing therapies focused on patient convenience and improved healthcare outcomes.

With increasing interest in personalized and patient-controlled treatments, etripamil may influence future directions in cardiovascular drug development.

References

1. FDA. **Cardamyst (Etripamil) Prescribing Information.**
2. Stambler BS et al. **Clinical evaluation of intranasal etripamil for PSVT treatment.**
3. RAPID Trial: **Phase III study of etripamil nasal spray in PSVT patients.**
4. Milestone Pharmaceuticals. **Clinical development and FDA approval of Cardamyst.**